



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1-AT3-TECHNICAL PRESENTATION

CSA0504 : Database Management Systems

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Question 1: ER Diagram for an Online Library System

Introduction

What is an ER Diagram?

- ER stands for Entity Relationship Diagram.
- It is used to design databases.
- It shows entities, attributes, and relationships.
- It helps organize data efficiently.

ER Diagram Components

Entities

Student

Book

Librarian

Issue

Attributes

Student ID

Book ID

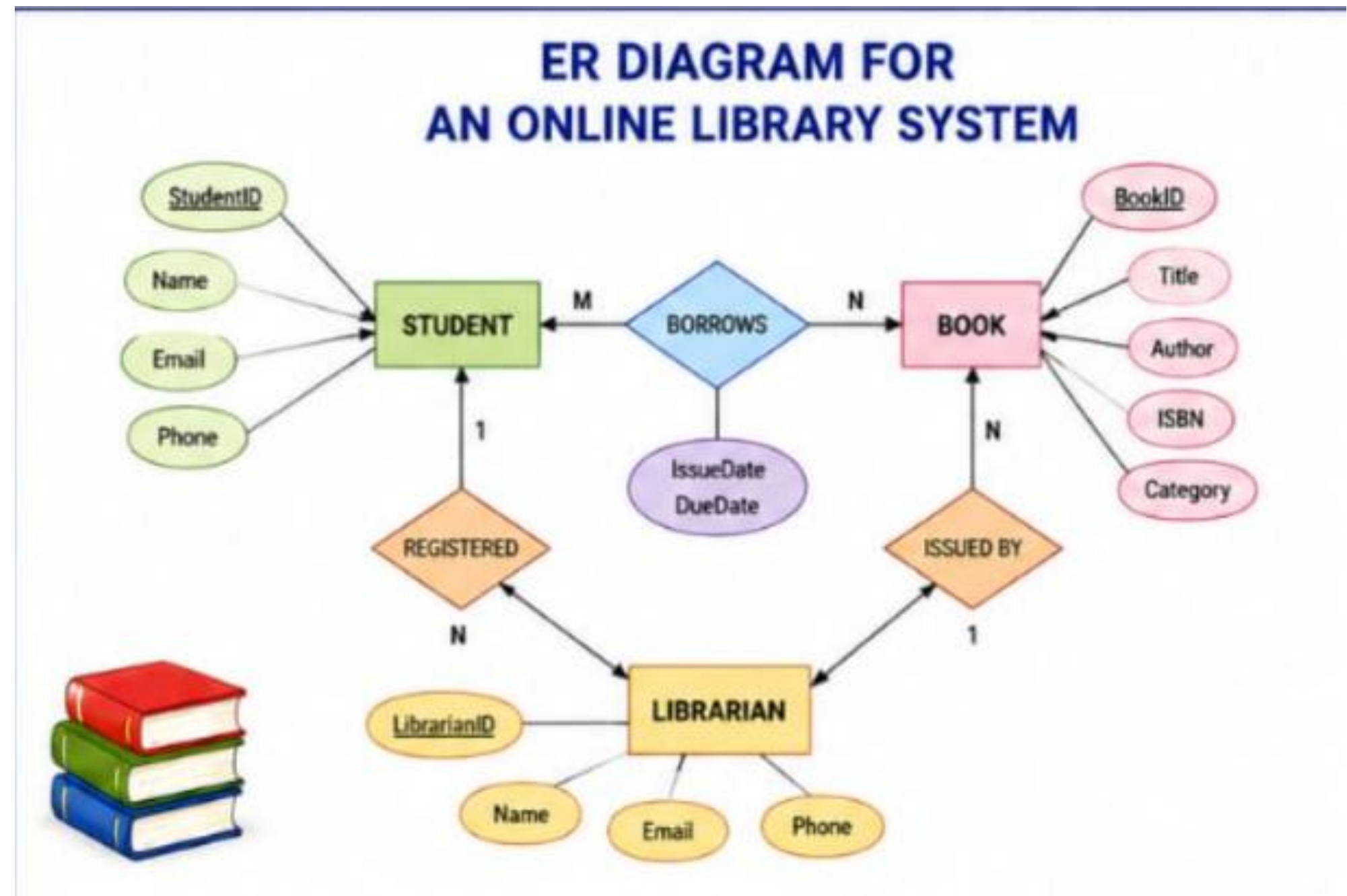
Book Title

Issue Date

Relationships

Student borrows Book.

Librarian issues Book.





Advantages & Conclusion

Advantages:

- Easy database design.
- Reduces data redundancy.
- Improves data organization.
- Easy maintenance.

Conclusion:

ER diagrams are the foundation of database design.



Question 2: EER Model – University Case Study

Introduction

- **Enhanced Entity Relationship (EER) Model** is an advanced version of ER (Entity-Relationship) model used for designing complex databases.
- It supports **inheritance**, allowing subclasses to inherit attributes from a superclass.
- **Generalization** combines similar entities into one higher-level entity.
- **Specialization** divides a general entity into specific sub-entities.
- It helps represent real-world systems more accurately and efficiently.

University Case Study

Superclass

- Person

Subclasses

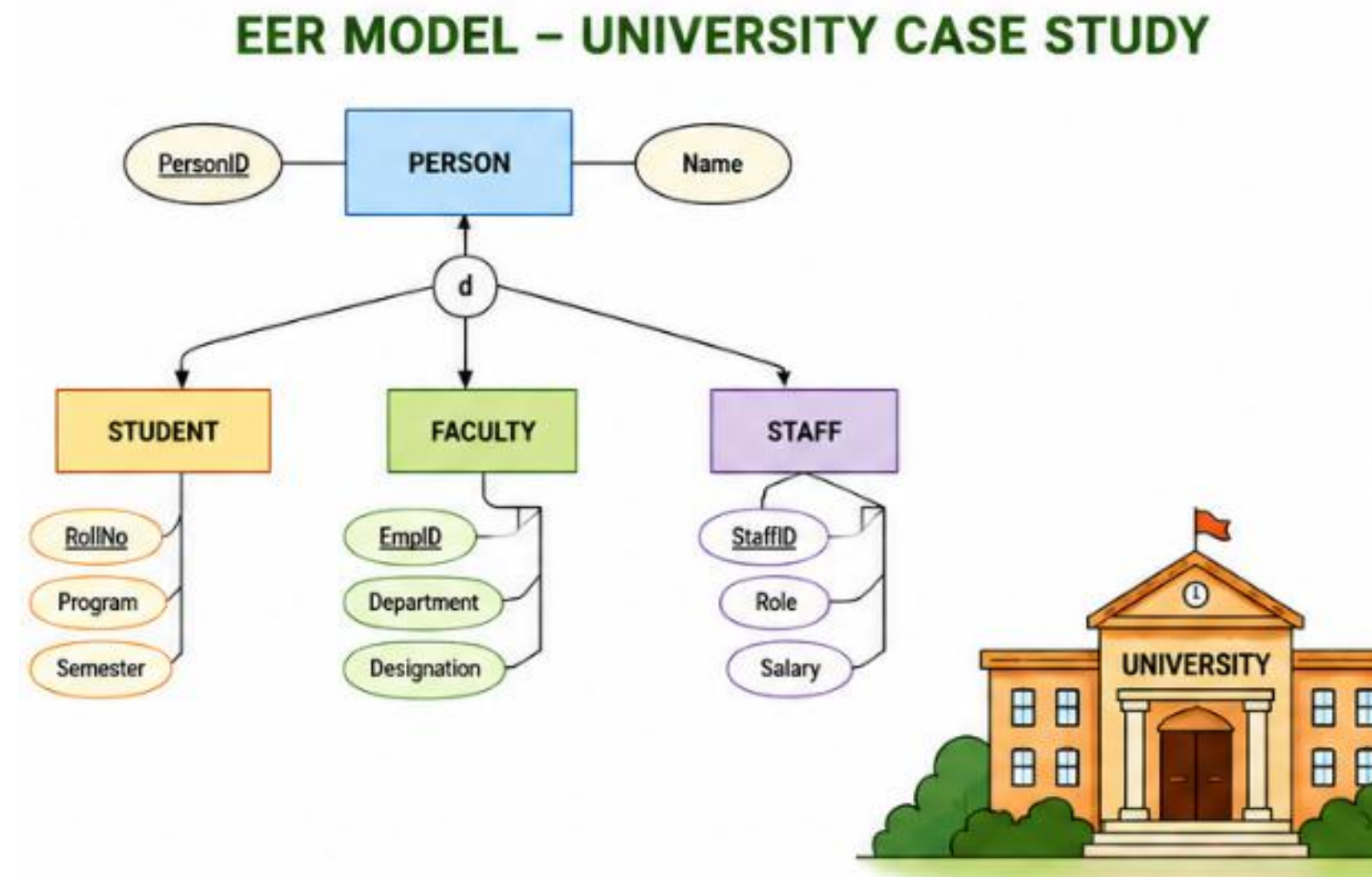
- Student
- Faculty
- Staff

Generalization

- Combines Student, Faculty, and Staff into Person.

Specialization

- Person is divided into Student, Faculty, and Staff.



Advantages & Conclusion

Advantages:

- Represents real-world scenarios.
- Supports inheritance.
- Improves database flexibility.
- Reduces redundancy.

Conclusion:

EER models make database design more effective.



Question 3: Data Independence in DBMS



Introduction

- **Data independence** allows changes to the database without affecting application programs.
- It makes the database **flexible** and easy to update.
- It **reduces maintenance effort** and saves time.
- It **improves performance** by enabling efficient data management.

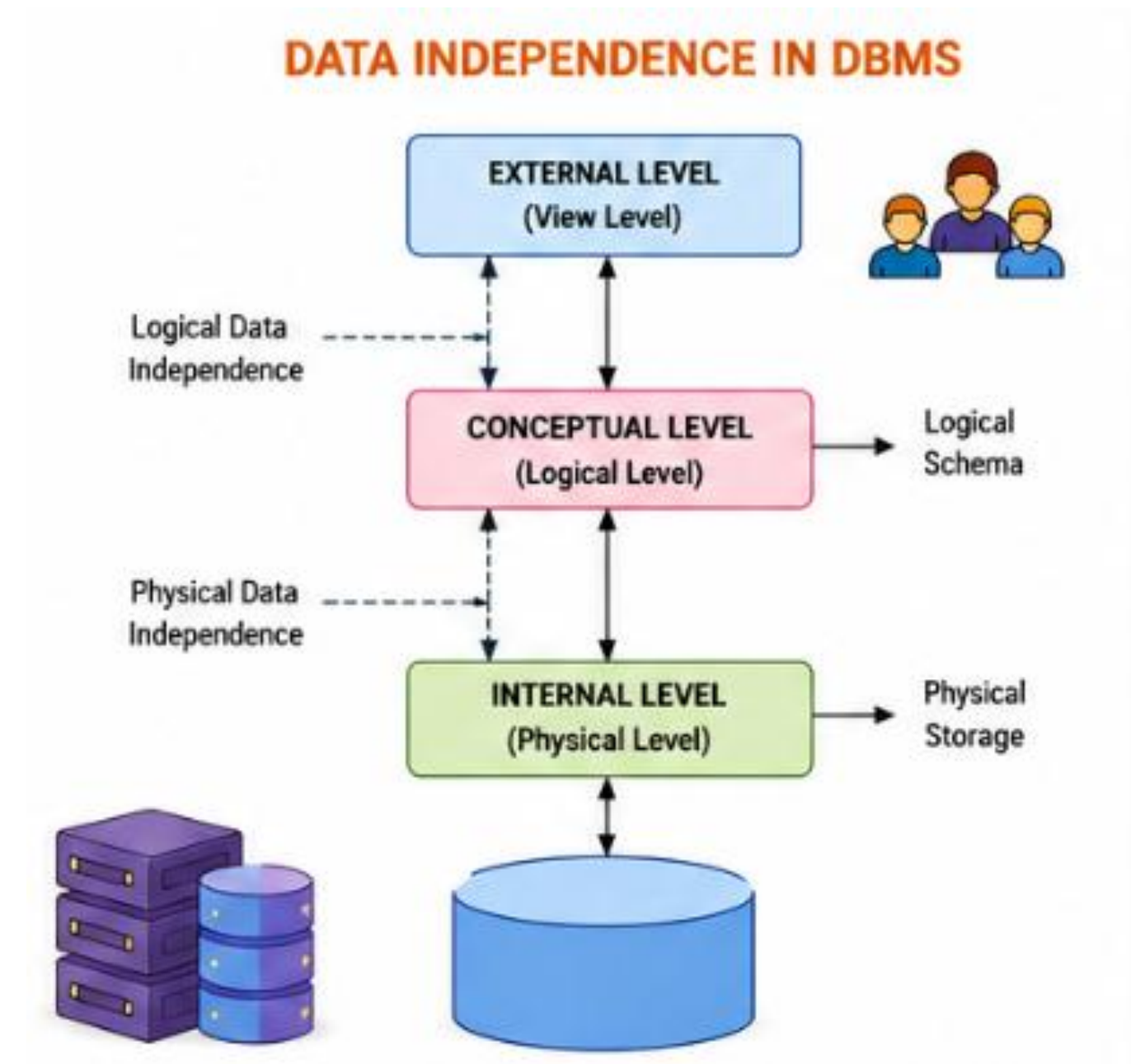
Types of Data Independence

Logical Data Independence

- Changes logical schema.
- No changes to applications.

Physical Data Independence

- Changes physical storage.
- Logical schema remains unchanged.



Advantages & Conclusion

Advantages:

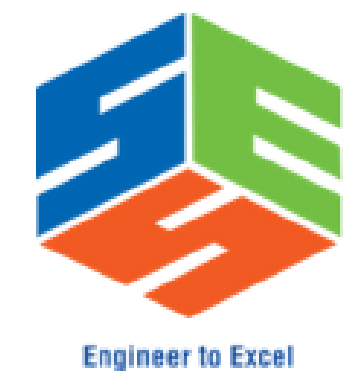
- Easy to modify the database structure.
- Reduces application maintenance.
- Improves database flexibility and scalability.
- Enhances overall system performance.

Conclusion:

Data independence enhances database efficiency by allowing changes without affecting application programs.



Question 4: Software Components of DBMS



Query Processor

Functions:

Parses SQL Queries: Checks the syntax and converts SQL statements into an internal format.

Optimizes Queries: Selects the most efficient execution plan to improve performance.

Executes Queries: Processes the optimized query and performs the requested operations.

Retrieves Required Data: Fetches the requested data from the database and returns the results to the user.

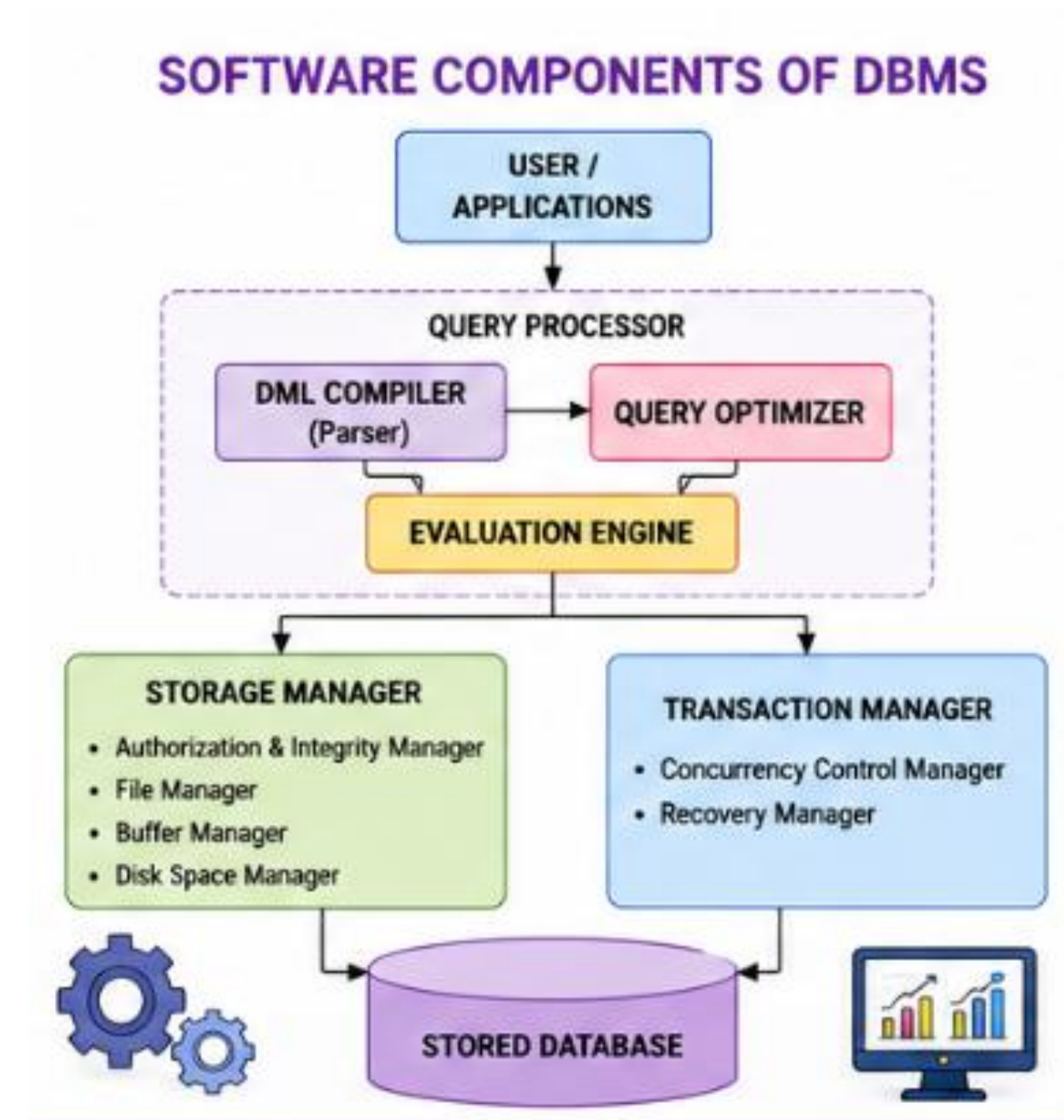
Storage Manager & Transaction Manager

Storage Manager

- Stores data.
- Manages files.
- Handles memory.

Transaction Manager

- Maintains ACID properties.
- Controls concurrent transactions.
- Performs recovery.





Conclusion

- The **Query Processor** interprets and executes SQL queries efficiently.
- The **Storage Manager** organizes, stores, and retrieves data securely.
- The **Transaction Manager** ensures data consistency and maintains ACID properties.
- **Together, these components ensure efficient, secure, and reliable database management.**



Question 5: Strong and Weak Entities



Strong Entity

Features

- Has a **Primary Key** to uniquely identify each record.
- Exists **independently** without relying on another entity.
- Can be **uniquely identified** in the database.

Examples

Student – Student ID

Employee – Employee ID

Customer – Customer ID



Weak Entity

Features

- **Has no Primary Key** and cannot be identified on its own.
- **Depends on a Strong Entity** for its existence.
- **Uses a Partial Key** along with the strong entity's key for identification.

Examples

Dependent – Depends on an Employee.

Order Item – Depends on an Order.

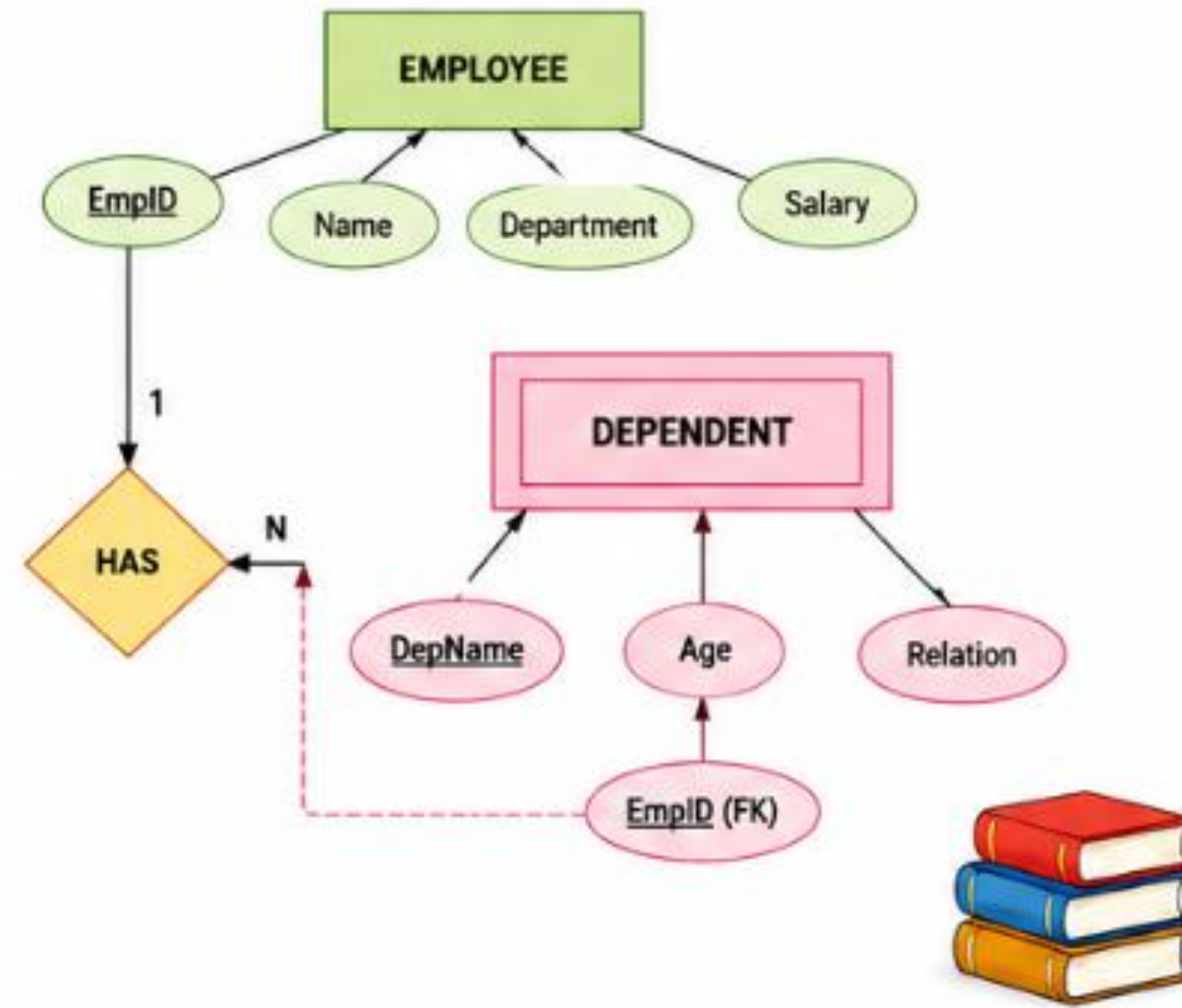
Book Copy – Depends on a Book.



Engineer to Excel



STRONG AND WEAK ENTITIES IN ER MODEL



Comparison & Conclusion

Strong Entity	Weak Entity
Has Primary Key	No Primary Key
Independent	Dependent
Own Existence	Depends on Strong Entity
Single Rectangle	Double Rectangle

Conclusion:

Strong and weak entities help model real-world relationships accurately and improve database design.

Thank
you
Love

